ALAHMEDS STUDY / MACROECONOMICS

Midterms 1-20, 28 | Formulas and graph rules

Coverage: 21 of 28 review modules. Generated 11 Nov 2026. Compact cues; full explanations on the website.

Prices and real output [M1]

$\pi = (P_{new}/P_{old} - 1) \times 100\%$

P price index; π inflation per period.

Nominal GDP = $\Sigma\, p_t q_t$

Real GDP = $\Sigma\, p_{base} q_t$

p price,q quantity; Σ sum; base prices fixed.

Fixed-basket index = $(cost_t/cost_{base}) \times 100$

Same fixed basket. Divide index by100 for ratio.

GDP deflator = (nominal GDP/real GDP) × 100

Real GDP=nominal/(deflator/100).

$g_{real} = (1+g_{nominal})/(1+\pi_{GDP}) - 1$

Decimal rates; approx. nominal minus inflation.

SRC-011 slides52-57;SRC-079.

GDP and saving [M2,5]

Y = C + I + G + NX; NX = EX − IM

C consumption,I capital,G purchases,NX exports−imports.

I: new fixed capital + homes + Δinventories

New capital+homes+inventory change. No transfers in G.

Value added = output − intermediate inputs

Count final output/value added/income once.

S = Y − C − G = I + NX; NX = S − I

S national saving; all flows per period.

SRC-011 slides34-50;SRC-081;012s13.

Production and growth [M4-5]

$Y = A \times K^{0.3} \times N^{0.7}$

Y output,K capital,N labor hours,A efficiency.

$ALP = Y/N = A(K/N)^{0.3}$

ALP output/hour. Fixed A,K:N↑→Y↑,ALP↓.

$Y'/Y = (A'/A)(K'/K)^{0.3}(N'/N)^{0.7}$

Prime=new; exact growth factors.

$g_Y \approx g_A + 0.3g_K + 0.7g_N$

g growth; small-change approximation; same periods.

$g_A \approx g_Y - 0.3g_K - 0.7g_N$

TFP residual; A differs from output/hour.

$g_{Y/N} \approx g_A + 0.3(g_K - g_N) \approx g_Y - g_N$

Approximate ALP growth; same period rates.

SRC-012 slides4-15;SRC-082.

Work reward and total hours [M7,10]

$w = W/P$; $R = [(1-t_n)/(1+t_c)]w$

W $/hour;P $/good;w,R goods/hour;tax rates decimal.

$N_s(w) = \Sigma_j h_j(w)$; equal h: N_s = pop×h

h hours/person; sum at the same wage.

$N_d(w^*) = N_s(w^*) = N^*$

Star=clearing; N total hours, not workers.

SRC-012 slides21-35;SRC-083.

Buying power and interest [M1]

Goods bought = dollars / (dollars per good)

Price per good in denominator.

$r^e \approx i - \pi^e$; $r^a \approx i - \pi^a$

i nominal;e expected;a actual. Match periods.

$1+r^a = (1+i)/(1+\pi^a)$

Exact real return. Surprise inflation hurts lenders.

SRC-011 slides59-62;SRC-129p2.

Capital and investment [M12]

$P \times MPK_{next} = p(k)(r+\delta)$

If P=p(k): $MPK_{next} = r+\delta$

Expected MPK;P output price,p(k) capital price;r,δ same period.

$K' = (1-\delta)K + I$; $I = K' - (1-\delta)K$

K stock,I gross flow,δ depreciation share.

Net I = I − δK = K′ − K

δK replacement; gross I>0 can still shrink K.

r ↓ → desired K′ ↑ → I ↑ (current K fixed)

r vertical,I horizontal. Expected A↑ shifts I demand→.

SRC-143 slides4-13;SRC-085.

Household budget [M13]

$c_f = (a+y-c)(1+r)+y_f$

a net assets(debt negative);y disposable real income;f future.

PVLR = $a+y+y_f/(1+r)$

PVLC = $c+c_f/(1+r)$ = PVLR

PVLC spending=PVLR resources, both valued today.

$MU(c) = \beta(1+r)MU(c_f)$

MU marginal utility;β future weight; identical utility.

$c = c_f = PVLR/[1+1/(1+r)]$

Equal smoothing if β(1+r)=1; free borrowing/lending.

s = y−c; $a_{next} = (a+s)(1+r)$

s saving flow;a_next net assets after interest.

SRC-143 slides17-22;SRC-086.

Deficits and debt [M18]

Taxes=t_nY; transfers=Tr−gY

t_n tax rate;g transfer withdrawal;Tr baseline term.

D_{actual}=G+Tr−(t_n+g)Y

Actual Y;G purchases;D deficit flow per period.

$D_{structural}$=G+Tr−(t_n+g)Y*

$D_{cyclical}$=(t_n+g)(Y*−Y)

Cyclical=actual−structural;recession raises D.

Debt/GDP = B/Y ×100%

B stock;Y annual GDP,same price basis;D flow. No adjustments:ΔB=D.

SRC-014 slides4-8,25.

Employment denominators [M3]

LF = E + U; pop = E + U + outside

E employed;U searching jobless;pop eligible16+ civilian.

Unemployment = U/LF × 100%

LF=E+U denominator; search exit can lower rate.

Participation = LF/pop × 100%

pop denominator, not price level P.

Employment/population = E/pop × 100%

People, not payroll jobs.

SRC-011 slides25-31.

Marginal products [M4,6]

$MPN = 0.7\, A(K/N)^{0.3} = 0.7Y/N$

MPN marginal output/hour, not average output.

$MPK = 0.3\, A(N/K)^{0.7} = 0.3Y/K$

Own input↑→own MP↓;K↑→MPN↑;N↑→MPK↑.

Competitive firm: MPN = w = W/P

w goods/hour. Hire while MPN>w.

Labor income = 0.7Y; capital income = 0.3Y

Competitive income shares; A,K↑ shift labor demand→.

SRC-012 slides4-7,18-20.

Potential output and graph axes [M11]

$Y^* = A\, K^{0.3}(N^*)^{0.7}$

Clearing hours N*;Y* not physical maximum.

Labor: W/P vertical; N horizontal

Demand down,supply up with lifetime resources fixed.

LRAS: P vertical; Y horizontal; Y = Y*

Flexible W:P,W double→same real w,N*,Y*.

A,K or clearing N* changes → LRAS can shift

P alone:along LRAS. Sticky W may put Y belowY*.

SRC-012 slides52-59;SRC-129p13.

Borrowing limits [M16]

$c_1^d = (a+y_1+y_2)/2$

r=0,β=1;equal periods,no bequest;a,y≥0;d desired.

No borrowing: $c_1 = \min(a+y_1, c_1^d)$

No borrowing; saving allowed. Rule may not bind.

Finite-change MPC = $(c_{after} - c_{before})/\Delta y_1$

Use actual plans. Large gains can cross regimes.

SRC-128;SRC-143slides28-29,31.

Capacity and crowding out [M20]

At Y*: Y*=C+I+G; NX=0

Hold A,K,N*,C fixed;NX=0;Y*=capacity benchmark.

ΔI=−ΔG; r rises to reduce desired I

Full crowding under these assumptions, not every recession.

SRC-014 slides18-21,25;SRC-134.

Labor changes: horizon matters [M7-10]

w ↑ → N_s ↑, PVLR fixed; PVLR ↑ → N_s ↓, w fixed

Normal leisure; movement vs wealth shift.

Temporary A ↑: N_d →; N*, w*, Y ↑

K fixed,wealth effect negligible. Leisure↑:N,Y↓;w↑.

Permanent A ↑: demand →; wealth effect on supply ←

Normal leisure; N ambiguous,w↑;course Y↑.

Temporary tax ↑: N_s ←; N* ↓; w* ↑; R ↓

A,K fixed; negligible wealth effect; no transfer.

Permanent tax ↑: substitution ←; income →

No offsetting transfer;normal leisure;net N ambiguous.

Population ↓ → aggregate N_s ←

Remaining individual schedules can stay unchanged.

Value of leisure ↑ → individual + aggregate supply ←

Demand fixed:N*↓,w*↑. Wage alone moves along supply.

SRC-012 slides21-39,54;SRC-083.

Income news and time [M14-15]

c = (a + Σ remaining y)/L

r=0,β=1; free borrowing; L equal remaining periods.

MPC = Δc(today)/Δy(today)

Dollar changes, specified experiment; not growth rates.

One-time Δy today: Δc = Δy/L; MPC=1/L

Every remaining period Δy: Δc=Δy; MPC=1

Surprises. One-time assets:Δc=Δa/L.

Replan: $c_{new} = [a_{now} + \Sigma \text{ remaining expected } y_{new}]/L_{left}$

Signed assets include debt; past consumption is over.

Keynesian $c = a_0 + b\, y$; MPC=b

a0 intercept,b MPC; different current-income model.

SRC-143 slides15,22-26;SRC-127.

IS: demand and interest [M17]

$Y_d = C + I(r) + G$; NX=0

r vertical,Y horizontal;closed;Y_d demanded,may≠Y*.

Illustration: $I = I_0 - br$

$Y_d = C+G+I_0 - br$

Generated:C,G fixed;b>0 rate sensitivity;r decimal.

SRC-014 slides10-16;SRC-087.

Spending multiplier [M19]

$k_G = \Delta Y/\Delta G$; $\Delta Y = k_G \Delta G$

Causal change vs no-policy baseline, not correlation.

Special illustration: $k_G = 1/(1-m)$

Special:0≤m<1 MPC;fixed r,slack;no tax/import leaks.

SRC-014 slide26;SRC-040.

Exam: one handwritten standard-size sheet, front only; no photocopy. This PDF is a study/hand-copy guide. [SRC-127]